Suriyan Dhanapal

Generative AI | Product Engineer | Backend, Systems & Infrastructure

Coimbatore, India | suriyandhanapal@gmail.com | +91 81100 44608
linkedin.com/in/suriyan-d-74514223a | github.com/iamsuriyan | iamsuriyan.github.io/Portfolio

SUMMARY

Generative AI product engineer with 2+ years of experience building and shipping LLM products end to end, from knowledge ingestion and retrieval to backend, frontend and deployment. Built a compliance agent that answers from official Indian regulatory sources (DGFT, GST, CBIC) using hybrid retrieval, a multi-tenant RAG platform on pgvector and Weaviate, and a unified connector layer across 25+ LLM, STT and TTS providers serving live voice-agent calls. Strong in Python/FastAPI backends, agent design with human-in-the-loop controls, and building reliable, auditable AI systems for production use.

TECHNICAL SKILLS

GenAI & LLM: LLM integration (OpenAI, Google Vertex AI/Gemini, AWS Bedrock, Groq, Sarvam), AI agents and agentic workflows, LangGraph, tool/function-calling schema design, prompt design, human-in-the-loop approval gates, provider regression testing

Retrieval & Knowledge Systems: RAG, hybrid retrieval (keyword + semantic), semantic search, chunking, embeddings, knowledge-base ingestion pipelines, metadata-scoped multi-tenant retrieval, pgvector, Weaviate

Speech & Multimodal: STT, TTS, STS pipelines (Whisper, ElevenLabs, Sarvam), real-time voice agents over WebRTC/SIP, vision-model document extraction, OCR

Backend & Data: Python, FastAPI, Node.js, Express, REST API design, microservices, gRPC, WebSockets, async workers (Redis/RQ), OAuth 2.0, JWT, RBAC, multi-tenancy, idempotency and retry design, webhook delivery, PostgreSQL, MongoDB, Redis, Cassandra, SQLAlchemy, SQL

Infrastructure & Tools: Docker, Linux/shell, nginx, GCP, AWS (S3, Bedrock), GitHub Actions (CI/CD), Git, Claude Code

Frontend: React, Next.js, TypeScript, Redux Toolkit, Vite, Tailwind CSS, React Native (Expo)

PROFESSIONAL EXPERIENCE

Product Engineer, **Cooper Compass** **Aug 2026 – Present**
IIT Bombay-incubated startup | Thavasi Scrutiny: Indian customs compliance platform | thavasi.coopercompass.com

Tech: Python, FastAPI, hybrid retrieval, OpenAI, vision-model OCR, pdfplumber, camelot, pytesseract, SQLAlchemy, PostgreSQL, Redis/RQ, JWT, Docker, nginx, React, TypeScript, Vite

- Built the platform end to end as the product engineer: architecture, knowledge ingestion, extraction pipeline, rule engine, backend, React frontend and deployment. Customs House Agents upload a draft checklist plus source documents and receive a field-by-field discrepancy report, with every value traced to the document, page and cell it was read from; piloted with real Customs House Agents against live filings.
- Developed a compliance agent that answers Indian trade and tax compliance questions directly from facts served by official regulatory sources, including DGFT, the GST Portal, CBIC and ICEGATE, using hybrid (keyword + semantic) retrieval so answers are grounded in retrieved source facts rather than model memory.
- Built the knowledge store behind the agent: pipelines that collect, clean, normalize, chunk and index regulatory content from these sources into a searchable store, tuned for hybrid retrieval over legal and procedural text.
- Built a cheapest-path-first hybrid extraction pipeline: pdfplumber, camelot, pytesseract and openpyxl do the reading, with a vision model gated to genuinely scanned pages so inference cost is spent only where deterministic parsing fails. Runs in Redis/RQ workers so uploads return immediately and the client polls for completion.

AI / Software Engineer, **Mindgryd Software Private Limited** **Jun 2025 – Aug 2026**
AI agent platform | Coimbatore, India

Tech: Python, FastAPI, LangGraph, RAG, Weaviate, pgvector, PostgreSQL, OpenAI, Vertex AI/Gemini, AWS Bedrock, Groq, Whisper, ElevenLabs, Sarvam, WebRTC, SIP, gRPC, Redis, MongoDB, Cassandra, Docker, GCP, Next.js, TypeScript

- Integrated 25+ LLM, STT, TTS and STS providers (OpenAI, Gemini, Vertex AI, Groq, Sarvam, Whisper, ElevenLabs) behind a single unified connector interface on FlowCat, the team's open-sourced real-time media engine, cutting new-provider integration from days to hours and serving ∼100 live voice-agent calls/day in pilot.

- Normalized provider divergence behind that interface (incompatible function-calling and tool-call schemas, gRPC streaming transports, API-key resolution) and built provider-aware regression flows that caught breaking vendor changes before release rather than on live calls.
- Built the multi-tenant RAG pipeline behind the AI memory and context platform: chunked and embedded ingested documents into PostgreSQL/pgvector with metadata-scoped retrieval, so every query stayed inside the requesting organization's own corpus, with tenant isolation enforced at the retrieval layer, not just the API.
- Built a workspace-aware agent recommendation system over semantic search: embedded onboarding context, indexed it in Weaviate, and retrieved and ranked from several hundred personalized agents inline on every search.
- Designed the agent autonomy model, human-in-the-loop approval gates governing how far an agent could act before an action required sign-off, plus the execution and schema design behind the workflow automation platform.
- Built the knowledge-ingestion layer feeding retrieval: 8 connectors including Google Drive, Dropbox, Notion and SharePoint, with per-user OAuth token storage and refresh and content normalization across heterogeneous source formats.
- Debugged provider-specific failures on live voice calls (streaming disconnects, tool-call mapping mismatches, key resolution) and built a real-time Vite + React dashboard for call status, provider health and test-flow results, which became the team's primary triage tool.
- Owned platform engineering: OAuth 2.0 authentication with Google/GitHub/Microsoft, JWT sessions, RBAC and automatic organization provisioning; an HMAC-signed, at-least-once webhook delivery service handling hundreds of events/day; and the Razorpay/PayPal payments layer.

Software Engineer, **Hifresh Agro Private Limited** **Sep 2024 – Jun 2025**
Agro-tech product company | Coimbatore, India
Tech: Python, FastAPI, Node.js, Express, REST APIs, MongoDB, React Native (Expo), Redux, Zoho Books API, response caching

- Built and owned a full-stack delivery-management application used daily by the delivery team and handling ∼150 orders/day: React Native/Expo client on an extended Node.js/Express backend, with APIs for delivery assignment, billing and Zoho Books invoicing.
- Added response caching and structured logging to the highest-traffic endpoints, cutting response time from ∼2 s to under ∼300 ms.
- Worked on a pilot project for a social media platform, building backend services and REST APIs in Python and FastAPI.

EDUCATION

Bachelor of Engineering, Electronics and Communication Engineering **2019 – 2023**
Dr. N.G.P. Institute of Technology, Coimbatore

COURSES & CERTIFICATIONS

- **AI Agents Course, Certificate of Completion**, Hugging Face: agentic workflows and tool calling **2025**
- **Building with the Claude API**, Anthropic Academy: tool use, RAG and agent architectures **2026**